

AP Physics C: Mechanics

The Complete Exam Guide

All 7 units explained with the calculus you need, the four free-response question types, a lab playbook, and 10 hard practice questions.

Hi, I'm **Pigo!** I'll drop tips all through this guide.



3 hrs

85 min multiple choice
+ 95 min free response

42+4

multiple-choice +
free-response questions

7 units

calculus-based
mechanics

72–73%

scored 3 or higher
in recent years

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01 THE EXAM AT A GLANCE

Two sections, one score

Your AP score (1–5) combines a multiple-choice section and a free-response section, each worth half. This is a calculus-based course: expect derivatives and integrals throughout. A calculator and an equation sheet are allowed on the whole exam.

50%

Section I • Multiple choice • 42 questions

50%

Section II • Free response • 4 questions

Exam structure

Section	Weight & timing	What you do
I • Multiple choice	50% of score 1 hr 25 min 42 questions	Single questions and short sets built around a stimulus such as a diagram, graph or data table. Calculator allowed.
II • Free response	50% of score 1 hr 35 min 4 questions	Four question types: Mathematical Routines • Translation Between Representations • Experimental Design and Analysis • Qualitative/Quantitative Translation.

HYBRID DIGITAL

Click in Bluebook, write on paper

You answer multiple-choice questions and read the free-response questions in the **Bluebook** app, then handwrite your free-response answers in a paper booklet. Practice writing clear, organized work by hand.

PROVIDED ON EXAM DAY

Equation sheet + calculator

A table of equations and constants is provided, and a calculator is allowed throughout. Knowing **when** an equation applies, and its conditions, matters more than memorizing it.

How to use this guide

Just starting?

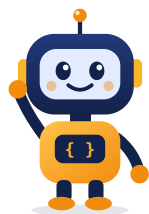
Read pages 2–6, then follow the roadmap on page 8 month by month.

Mid-year?

Check the traps on page 7 and drill the free-response playbook on page 6.

Exam in weeks?

Take the 10 questions on page 9 in 22 minutes, then use the final-four-weeks plan.



Pigo's tip: Physics answers are scored on **reasoning**. Start every explanation from a principle, such as Newton's second law or conservation of energy, then apply it to the situation.

SECTION I

02 The multiple-choice section

42 questions in 85 minutes gives you about 2 minutes per question. Many questions test reasoning with symbols and graphs rather than plugging in numbers.

42

questions, some in stimulus-based sets

~2:00

minutes per question on average

50%

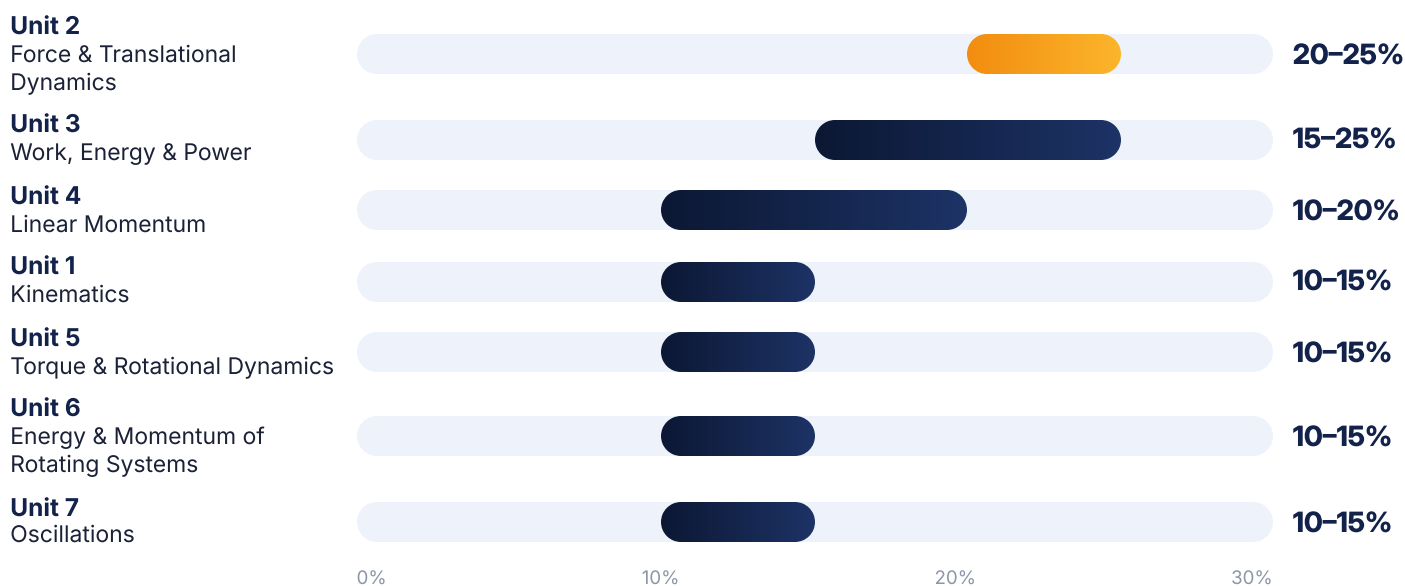
of your total AP score

0

points lost for a wrong answer, so never leave one blank

How the questions are weighted by unit

College Board gives a range for each unit's share of the multiple-choice section. Bars show minimum to maximum.



What this means for your study time: Forces and energy carry the most weight, but rotation (Units 5–6) together is as large as either. Calculus shows up in every unit.

Strategies that work on this section

- **Draw a diagram** for every question: forces, fields, circuits or rays. It takes seconds and prevents sign errors.
- **Check limiting cases.** What happens if a mass is zero or an angle is 90° ? Wrong choices often fail this test.
- **Use units to eliminate** answer choices with the wrong dimensions.
- **Read graphs as relationships:** slope and area under the curve both have physical meaning.
- **Work symbolically first**, then plug in numbers only if needed.
- **Flag and move on** after 3 minutes. Return once the quick points are banked.

03 WHAT THE COURSE COVERS

The 7 units, in plain English

Everything on the exam fits into these units.

UNIT

1

10–15%

Kinematics

Be ready to: differentiate and integrate position, velocity and acceleration; handle non-constant acceleration; analyze 2D and projectile motion; use relative velocity.

UNIT

2

20–25%

Force & Translational Dynamics

Be ready to: apply Newton's laws to systems; handle velocity-dependent drag with differential equations; analyze circular motion and gravitation; find the center of mass.

UNIT

3

15–25%

Work, Energy & Power

Be ready to: calculate work as $\int \mathbf{F} \cdot d\mathbf{x}$; relate force and potential energy ($F = -dU/dx$); use conservation of energy; calculate power as dW/dt .

UNIT

4

10–20%

Linear Momentum

Be ready to: use impulse as $\int \mathbf{F} dt$; apply conservation of momentum in collisions and explosions; track center-of-mass motion.

UNIT

5

10–15%

Torque & Rotational Dynamics

Be ready to: calculate rotational inertia by integration and the parallel-axis theorem; apply $\Sigma \tau = I\alpha$; analyze static equilibrium.

UNIT

6

10–15%

Energy & Momentum of Rotating Systems

Be ready to: use rotational kinetic energy and rolling without slipping; apply conservation of angular momentum; analyze orbits.

UNIT

7

10–15%

Oscillations

Be ready to: derive and solve the SHM differential equation; find periods for springs and physical pendulums; track energy in oscillators.

Calculus is expected. You should be comfortable with derivatives, integrals and simple differential equations, ideally from AP Calculus taken before or alongside this course.

04 SECTION II

The 4 free-response questions

95 minutes, 4 questions. The four question types are the same every year, so you can practice each on purpose.

MR

TBR

LAB

QQT

Mathematical Routines and Lab questions are typically worth the most points

MR

Mathematical Routines

Derive and calculate

Use equations and principles to derive expressions and calculate quantities, often starting symbolically.

Start from a fundamental equation and show each substitution.

TBR

Translation Between Representations

Diagrams, graphs, equations

Draw a free-body diagram, energy bar chart or graph, then write equations that are consistent with it.

Arrows on free-body diagrams start on the object and are labeled.

LAB

Experimental Design & Analysis

Plan and analyze

Design a procedure, choose what to graph to get a straight line (linearize), then use the slope to find a quantity.

Linearize: graph the quantities that make $y = mx + a$ a straight line.

QQT

Qualitative/Quantitative Translation

Explain, then verify

Explain a situation in words with physics reasoning, then derive an equation that agrees with your explanation.

Make sure your words and your equation predict the same trend.

◆ A time plan for the 95 minutes

MR

~25 min

TBR

~22 min

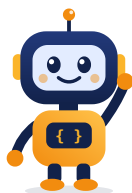
LAB

~25 min

QQT

~23 min

A suggested split, not a rule. Start with the question type you're strongest at, and never leave a question blank.



Points are given for **derivations**. "Plugging in" without a starting equation often loses credit.

FREE-RESPONSE PLAYBOOK

05

Anatomy of a full-credit answer

Here is a short practice prompt with a full-credit answer, and the habits that earn points.

PRACTICE TASK • WRITTEN BY ACT SAT GO

Practice prompt (MR style): A 2.0 kg object is dropped from rest and experiences air resistance $F = -bv$, with $b = 0.50 \text{ kg/s}$. (a) Write Newton's second law as a differential equation for $v(t)$. (b) Derive the terminal speed. (c) Solve for $v(t)$.

Full-credit response

- ① (a) Taking down as positive: $m \, dv/dt = mg - bv$.
- ② (b) At terminal speed, $dv/dt = 0$, so $v_T = mg/b = (2.0)(9.8)/0.50 = \mathbf{39.2 \text{ m/s}}$.
- ③ (c) Separate: $dv/(mg - bv) = dt/m$. Integrate from 0 to v and 0 to t : $-(1/b) \ln[(mg - bv)/mg] = t/m$.
- ④ $v(t) = (mg/b)(1 - e^{-bt/m}) = 39.2(1 - e^{-0.25t}) \text{ m/s}$.

- ① **Choose a sign convention** and write $\Sigma F = ma$ with every force.
- ② **Use the physics condition** (zero acceleration) before any calculus.
- ③ **Separate variables** and show the limits of integration.
- ④ **Check the result:** $v(0) = 0$ and $v \rightarrow v_T$ as $t \rightarrow \infty$.

Five habits for every free-response question

1

Answer every lettered part, in order, with a label.

2

Show the equation, the substitution and the units.

3

Justify with a cause $\rightarrow$ effect sentence, not a word.

4

Use the reference sheet; you don't need to memorize it.

5

Blank part? Write the relationship you would use. It can earn credit.

06 LEARN FROM OTHERS

10 traps that cost points

Meet **Glitch**. He lives in these ten mistakes. Squash each one before May.

$F = ma$ works for variable mass too! (Careful.)



01 Constant-acceleration equations everywhere

If acceleration changes with time or position, integrate instead of using the kinematic equations.

02 Missing the chain rule

For $a(x)$, use $a = v \, dv/dx$. Many students get stuck trying to integrate with respect to time.

03 Wrong axis for I

Rotational inertia depends on the axis. Use the parallel-axis theorem: $I = I_{cm} + Md^2$.

04 Sign of potential energy

$F = -dU/dx$. Forces point "downhill" on a $U(x)$ graph.

05 Rolling friction does work

In rolling without slipping, static friction does no work on the object.

06 Angular momentum of a point

A particle moving in a straight line can have angular momentum about a point: $L = r \times p$.

07 Centripetal force on diagrams

Never draw " F_c ". A real force (tension, gravity, normal) provides it.

08 Forgetting limits of integration

Use definite integrals with physical limits, or solve for the constant with an initial condition.

09 Physical pendulum period

$T = 2\pi\sqrt{I/(mgd)}$, where d is the distance from the pivot to the center of mass.

10 Plugging in too early

Derive symbolically first. Many questions ask for answers "in terms of" given quantities.

Quick self-check: a uniform rod of length L pivots about one end. What is its rotational inertia? ($I = ML^2/12 + M(L/2)^2 = ML^2/3$.)

07

PLAN YOUR YEAR

Your study roadmap to May

A steady plan beats a last-minute rush. Shift the months to match your teacher's pacing, and keep doing a few problems from older units every week.



Sep – Oct Motion + forces	○	Units 1–2 biggest unit Calculus-based kinematics, Newton's laws, drag and circular motion. Practice setting up differential equations.
Nov – Dec Conservation	●	Units 3–4 Work integrals, potential energy curves, momentum and center of mass. Start one free-response question per week.
Jan – Feb Rotation	○	Units 5–6 Rotational inertia by integration, torque, rolling and angular momentum.
Mar – Apr Oscillations + review	●	Unit 7 + timed practice SHM, physical pendulums and energy. Begin full timed sections in April.
Early May Sharpen	○	Free-response drills All four question types under timing, with a focus on derivations.
Exam day May	●	AP Physics C: Mechanics exam Sleep well, pack your calculator, and use the checklist at the end of this guide.

• The final four weeks

Week	Focus	Target
4 weeks out	Diagnose: 21 multiple-choice questions plus 2 free-response questions	List your two weakest units
3 weeks out	Fix weak units; one of each free-response type	80%+ on unit drills
2 weeks out	Full 3-hour practice exam	Finish both sections on time
Final week	Light review: equation sheet, diagram rules, traps list	Rest the night before

08 TEST YOURSELF 10 practice questions

Original questions written by ACT SAT GO in the style of the exam, pitched at the harder end. Use $g = 9.8 \text{ m/s}^2$ unless stated. Give yourself **22 minutes** for all ten, then check page 10.

1 UNIT 1 • VARIABLE ACCELERATION

A particle starts at $x = 0$ with velocity 2 m/s , and its acceleration is $a = 6t \text{ (m/s}^2\text{)}$. What is its position at $t = 2 \text{ s}$?

- A 8 m B 10 m
C 12 m D 16 m

2 UNIT 2 • DRAG

A 2.0 kg object falls with drag force $F = -bv$, $b = 0.50 \text{ kg/s}$. How long does it take to reach about 63% of terminal speed?

- A 0.25 s B 2.0 s
C 4.0 s D 39 s

3 UNIT 3 • VARIABLE FORCE

A force $F(x) = 3x^2 \text{ N}$ acts along the x -axis. How much work does it do as an object moves from $x = 0$ to $x = 2 \text{ m}$?

- A 6 J B 8 J
C 12 J D 24 J

4 UNIT 3 • POTENTIAL ENERGY

A particle's potential energy is $U(x) = 2x^3 - 6x \text{ (J)}$. At $x = 2 \text{ m}$, the force on it is:

- A 18 N in the $-x$ direction
B 18 N in the $+x$ direction
C 4 N in the $-x$ direction
D 0 N

5 UNIT 6 • ANGULAR MOMENTUM

A 4.0 kg disk (radius 0.50 m) spins at 10 rad/s . A 1.0 kg lump of clay is dropped onto its rim and sticks. What is the new angular speed?

- A 5.0 rad/s B 6.7 rad/s
C 8.0 rad/s D 10 rad/s

6 UNIT 5 • ROTATIONAL DYNAMICS

A uniform rod of length 1.5 m is held horizontally, pivoted at one end, and released. What is its angular acceleration just after release?

- A 4.9 rad/s^2 B 6.5 rad/s^2
C 9.8 rad/s^2 D 19.6 rad/s^2

7 UNIT 7 • SHM

A 0.50 kg mass on a spring with $k = 200 \text{ N/m}$ oscillates on a frictionless surface. What is its period?

- A 0.10 s B 0.31 s
C 0.63 s D 3.1 s

8 UNIT 4 • IMPULSE

A force $F(t) = 0.2t^2 \text{ N}$ acts on a 0.50 kg object at rest from $t = 0$ to $t = 3 \text{ s}$. What is the object's final speed?

- A 1.2 m/s B 1.8 m/s
C 3.6 m/s D 5.4 m/s

9 UNIT 6 • ROLLING

A solid sphere ($I = \frac{3}{2}MR^2$) rolls without slipping from rest down a ramp, dropping 1.4 m vertically. What is its speed at the bottom?

- A 3.7 m/s B 4.4 m/s
C 5.2 m/s D 6.2 m/s

10 UNIT 2 • GRAVITATION

Satellite X orbits a planet at radius R . Satellite Y orbits the same planet at radius $4R$. What is the ratio of their orbital periods, T_Y/T_X ?

- A 2 B 4
C 8 D 16

09

CHECK YOUR WORK

Answer key & explanations

Read every explanation, even for questions you got right. Each one points to a pattern that keeps coming back on the exam.

#	Answer	Why
1	C • 12 m	$v = \int 6t \, dt = 3t^2 + 2$. $x = \int v \, dt = t^3 + 2t$. At $t = 2$: $8 + 4 = 12$ m. Using $x = \frac{1}{2}at^2$ is wrong because a isn't constant.
2	C • 4.0 s	$v(t) = v_T(1 - e^{-bt/m})$. $63\% = 1 - 1/e$, reached when $t = m/b = 2.0/0.50 = 4.0$ s (one time constant).
3	B • 8 J	$W = \int_0^2 3x^2 \, dx = x^3$ from 0 to 2 = 8 J.
4	A	$F = -dU/dx = -(6x^2 - 6) = -(24 - 6) = -18$ N, so 18 N in the $-x$ direction.
5	B • 6.7 rad/s	$I_{\text{disk}} = \frac{1}{2}(4)(0.25) = 0.50$; clay adds $(1)(0.25) = 0.25$. L conserved: $0.50 \times 10 = 0.75\omega$, so $\omega \approx 6.7$ rad/s.
6	C • 9.8 rad/s ²	$\tau = Mg(L/2)$, $I = ML^2/3$. $\alpha = \tau/I = 3g/(2L) = 3(9.8)/3 = 9.8$ rad/s ² .
7	B • 0.31 s	$T = 2\pi\sqrt{(m/k)} = 2\pi\sqrt{(0.0025)} \approx 0.31$ s.
8	C • 3.6 m/s	Impulse $J = \int_0^3 0.2t^2 \, dt = 0.2 \times 9 = 1.8$ N·s. $v = J/m = 1.8/0.50 = 3.6$ m/s.
9	B • 4.4 m/s	$mgh = \frac{1}{2}mv^2 + \frac{1}{2}(\frac{1}{2}mR^2)(v/R)^2 = 0.7mv^2$. $v = \sqrt{(gh/0.7)} = \sqrt{19.6} \approx 4.4$ m/s.
10	C • 8	Kepler's third law: $T^2 \propto r^3$. $T_Y/T_X = (4)^{3/2} = 8$.

9–10

Strong. Move on to full timed free-response practice.

6–8

On track. Revisit the unit behind every miss on page 4.

0–5

Build foundations now. Review calculus-based kinematics and Newton's laws (Units 1–2).

10 KNOW THE TARGET

How students score

In recent years, about 72–73% of students earned a 3 or higher on AP Physics C: Mechanics, and about 20–22% earned a 5. Steady preparation is what moves a score up a full point.

72–73%

of students earned a 3 or higher in recent years

20–22%

earned the top score of 5 in recent years

3+

is the score many colleges accept for credit or placement

What each AP score means

5

Extremely well qualified

4

Very well qualified

3

Qualified

2

Possibly qualified

1

No recommendation

Ranges reflect recent College Board score distributions and shift a little each year. Credit and placement policies vary by college, so check each college's AP policy.

Exam-day checklist

- ☐ **Bluebook** is installed and updated, and your device passed the exam check
- ☐ Device **fully charged**, plus **two pencils and a dark-ink pen** for the free-response booklet
- ☐ Admission ticket and photo ID, as your school requires
- ☐ Confirm your **exam day and start time** in May with your AP coordinator
- ☐ An **approved calculator** with fresh batteries
- ☐ Plan your time: about **2 min** per multiple-choice question and **23 min** per free-response question
- ☐ Answer **every** multiple-choice question. Wrong answers don't lose points
- ☐ A **ruler or straightedge** for graphs and diagrams, if permitted

In the free-response section: start from a fundamental principle, draw clean labeled diagrams, keep words and equations consistent, and show every step of a derivation before plugging in numbers.



ACT SAT GO • 1-ON-1 AP TUTORING

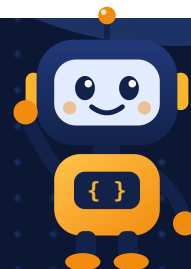
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